

CHAPTER 6

$$\text{P.P.6.1} \quad v = \frac{q}{C} = \frac{120 \times 10^{-6}}{3 \times 10^{-6}} = \underline{\underline{40\text{V}}}$$

$$w = \frac{1}{2} C v^2 = \frac{1}{2} \times 3 \times 10^{-6} \times 1600 = \underline{\underline{2.4\text{mJ}}}$$

$$\begin{aligned} \text{P.P.6.2} \quad i(t) &= C \frac{dv}{dt} = 10 \times 10^{-6} \frac{d}{dt} (50 \sin 2000t) \\ &= \underline{\underline{\cos 2000t \text{ A}}} \end{aligned}$$

$$\begin{aligned} \text{P.P.6.3} \quad v &= \frac{1}{C} \int_0^t i dt = \frac{10^{-3}}{0.1 \times 10^{-3}} \int_0^t 50 \sin 120\pi t \, dt \, \text{V} \\ &= -\frac{500}{120\pi} \cos 120\pi t \Big|_0^t = \frac{50}{12\pi} (1 - \cos 120\pi t) \text{V} \end{aligned}$$

$$v(t = 1\text{ms}) = \frac{50}{12\pi} (1 - \cos 0.12\pi) = \underline{\underline{93.14\text{mV}}}$$

$$v(t = 5\text{ms}) = \frac{50}{12\pi} (1 - \cos 0.6\pi) = \underline{\underline{1.7361 \text{ V}}}$$

$$\text{P.P.6.4} \quad i(t) = \begin{cases} 50t, & 0 < t < 2 \\ 100, & 2 < t < 6 \end{cases}$$

$$v = \frac{1}{C} \int i dt = \frac{1}{10^{-3}} \int i dt \cdot 10^{-3} = \int i dt$$

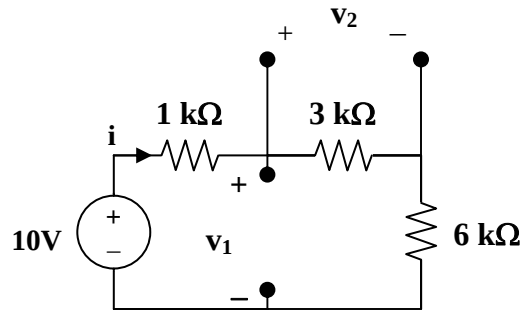
$$\text{For } 0 < t < 2, \quad v = \frac{1}{C} \int_0^t 50t \, dt = 25t^2 \times 10^3$$

$$\begin{aligned} \text{For } 2 < t < 6, \quad v &= \frac{1}{C} \int_2^t 100 dt + v(2) = (100t - 0.2 + 0.1) \\ &= (100t - 0.1) \text{V} \end{aligned}$$

$$\text{At } t = 2\text{ms}, \quad v = \underline{\underline{100\text{mV}}}$$

$$\begin{aligned} \text{At } t = 5\text{ms}, \quad v &= (500 - 100) \text{mV} \\ &= \underline{\underline{400 \text{ mV}}} \end{aligned}$$

P.P.6.5 Under dc conditions, the capacitors act like open-circuits as shown below:



$$i = \frac{10}{1 + 3 + 6} = 1\text{mA}$$

$$v_1 = (3\text{k} + 6\text{k})i = 9\text{V}$$

$$v_2 = (3\text{k})i = 3\text{V}$$

$$w_1 = \frac{1}{2}C_1v_1^2 = \frac{1}{2}(10 \times 10^{-6})(9)^2 = \underline{405\mu\text{J}}$$

$$w_2 = \frac{1}{2}C_2v_2^2 = \frac{1}{2}(20 \times 10^{-6})(3)^2 = \underline{90\mu\text{J}}$$

P.P.6.6 Combining 60 and 120 μF in series = $\frac{60 \times 120}{180} = 40\mu\text{F}$

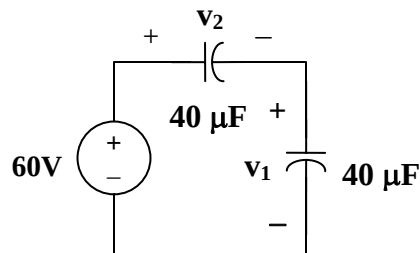
40 μF in parallel with 20 μF = 40 + 20 = 60 μF

50 μF in parallel with 70 μF = 50 + 70 = 120 μF

60 μF in series with 120 μF = $\frac{60 \times 120}{180} = \underline{40\mu\text{F}}$

P.P.6.7 60 μF in series with 30 μF = $\frac{60 \times 30}{90} = 20\mu\text{F}$

20 μF in parallel with 20 μF = 40 μF



From the Figure, $v_1 = v_2 = \frac{60}{2} = 30\text{V}$

Hence $v_2 = 10V_1$, $v_4 = 20\text{V}$

Note that $q_3 = q_4 = 60 \times 10\mu\text{C}$. Thus $v_1 = v_2 = \underline{30\text{V}}$, $v_3 = \underline{10\text{V}}$, $v_4 = \underline{20\text{V}}$

$$\begin{aligned}\text{P.P.6.8} \quad v &= L \frac{di}{dt} = 10^{-3} \frac{d}{dt} (20 \cos 100t) \cdot 10^{-3} \\ &= \underline{-2 \sin 100t \text{ mV}}\end{aligned}$$

$$\begin{aligned}w &= \frac{1}{2} Li^2 = \frac{1}{2} \times 10^{-3} (400 \cos^2 100t) \cdot 10^{-6} \\ &= \underline{200 \cos^2 100t \text{ nJ}}\end{aligned}$$

$$\begin{aligned}\text{P.P.6.9} \quad i &= \frac{1}{L} \int_{t_0}^t v(t) dt + i(t_0) = \frac{1}{2} \int_0^t 10(1-t) dt + 2 \\ &= 5 \left(t - \frac{t^2}{2} \right) + 2\end{aligned}$$

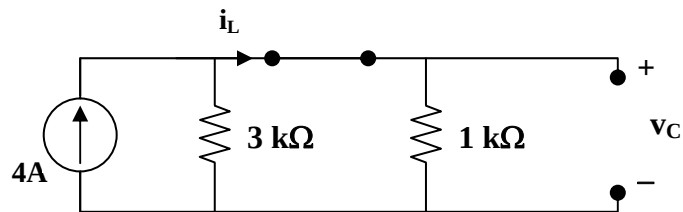
$$\text{At } t = 4, i = 5(4 - 8) + 2 = \underline{-18\text{A}}$$

$$\begin{aligned}p &= vi = 10(1-t) \left[5t - \frac{5}{2}t^2 + 2 \right] \\ &= 20 + 30t - 75t^2 + 25t^3\end{aligned}$$

$$\begin{aligned}w &= \int_0^4 p dt = [20t + 15t^2 - 25t^3 + 25t^4/4] \Big|_0^4 \\ &= 80 + 15 \times 16 - 1600 + 1600\end{aligned}$$

$$w = \underline{320\text{J}}$$

P.P.6.10 Under dc conditions, the circuit is equivalent to that shown below



$$i_L = \frac{3}{1+3} (4) = \underline{3\text{A}}$$

$$v_C = 1i_C = \underline{3\text{V}}$$

$$w_L = \frac{1}{2} Li_L^2 = \frac{1}{2} \left(\frac{1}{4} \right) (3)^2 = \underline{1.125\text{J}}$$

$$w_C = \frac{1}{2} Cv_C^2 = \frac{1}{2} (2)(3)^2 = \underline{9\text{J}}$$

P.P.6.11 40mH in series with 20mH = 40 + 20 = 60mH
 60mH in parallel with 30mH = 30 x 60/(90) = 20mH
 20mH in series with 100mH = 120mH
 120mH in parallel with 40mH = 40 x 120/(160) = 30mH
 30mH in series with 20mH = 50mH
 50mH in parallel with 50mH = 25mH

$$L_{eq} = \underline{\underline{25\text{mH}}}$$

P.P.6.12 (a) $i_2 = i - i_1 \longrightarrow i_2(0) = i(0) - i_1(0) = 1.4 - 0.6 = \underline{\underline{800 \text{ mA}}}$

$$(b) v_1 = 6 \frac{di_1}{dt} = 6(0.6)(-2)e^{-2t} = -7.2e^{-2t}$$

$$i_2 = \frac{1}{3} \int_0^t v_1 dt + i_2(0) = \frac{1}{3} \frac{(-7.2)}{(-2)} e^{-2t} \Big|_0^t + 0.8$$

$$= \underline{\underline{-0.4 + 1.2e^{-2t} \text{ A}}}$$

$$i = i_1 + i_2 = \underline{\underline{0.4 + 1.8e^{-2t} \text{ A}}}$$

(c) From (b),

$$v_1 = \underline{\underline{-7.2e^{-2t} \text{ V}}}$$

$$v_2 = 8 \frac{di}{dt} = 8(-2)(1.8)e^{-2t} = \underline{\underline{-28.8e^{-2t} \text{ V}}}$$

$$v = v_1 + v_2 = \underline{\underline{-36e^{-2t} \text{ V}}}$$

P.P.6.13 $RC = 25 \times 10^3 \times 10 \times 10^{-6} = 0.25$

$$v_o = -\frac{1}{RC} \int_0^t v_i(t) dt + v_o(0) = -\frac{1}{0.25} \int_0^t 10 dt \text{ mV} + 0$$

$$= \underline{\underline{40t \text{ mV}}}$$

P.P.6.14 $RC = 10 \times 10^3 \times 2 \times 10^{-6} = 2 \times 10^{-2}$

$$v_o = -RC \frac{dv_i}{dt} = -2 \times 10^{-2} \frac{d}{dt}(3t)$$

$$v_o = \underline{\underline{-60 \text{ mV}}}$$

P.P.6.15 $\frac{dv_o^2}{dt^2} = 4 \cos 10t - 3 \frac{dv_o}{dt} - 2v_o$

Using this we obtain the analog computer as shown below. We may let $RC = 1\text{s}$.

